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## This file is a general .xdc for the Basys3 rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top level signal names in the project

## Clock signal
set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports clk]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]


## Switches
#set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {sw[0]}]
#set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports {sw[1]}]
#set_property -dict { PACKAGE_PIN W16   IOSTANDARD LVCMOS33 } [get_ports {sw[2]}]
#set_property -dict { PACKAGE_PIN W17   IOSTANDARD LVCMOS33 } [get_ports {sw[3]}]
#set_property -dict { PACKAGE_PIN W15   IOSTANDARD LVCMOS33 } [get_ports {sw[4]}]
#set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports {sw[5]}]
#set_property -dict { PACKAGE_PIN W14   IOSTANDARD LVCMOS33 } [get_ports {sw[6]}]
#set_property -dict { PACKAGE_PIN W13   IOSTANDARD LVCMOS33 } [get_ports {sw[7]}]
#set_property -dict { PACKAGE_PIN V2    IOSTANDARD LVCMOS33 } [get_ports {sw[8]}]
#set_property -dict { PACKAGE_PIN T3    IOSTANDARD LVCMOS33 } [get_ports {sw[9]}]
#set_property -dict { PACKAGE_PIN T2    IOSTANDARD LVCMOS33 } [get_ports {sw[10]}]
#set_property -dict { PACKAGE_PIN R3    IOSTANDARD LVCMOS33 } [get_ports {sw[11]}]
#set_property -dict { PACKAGE_PIN W2    IOSTANDARD LVCMOS33 } [get_ports {sw[12]}]
#set_property -dict { PACKAGE_PIN U1    IOSTANDARD LVCMOS33 } [get_ports {sw[13]}]
#set_property -dict { PACKAGE_PIN T1    IOSTANDARD LVCMOS33 } [get_ports {sw[14]}]
#set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {sw[15]}]


## LEDs
set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports clean]
#set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports {led[1]}]
#set_property -dict { PACKAGE_PIN U19   IOSTANDARD LVCMOS33 } [get_ports {led[2]}]
#set_property -dict { PACKAGE_PIN V19   IOSTANDARD LVCMOS33 } [get_ports {led[3]}]
#set_property -dict { PACKAGE_PIN W18   IOSTANDARD LVCMOS33 } [get_ports {led[4]}]
#set_property -dict { PACKAGE_PIN U15   IOSTANDARD LVCMOS33 } [get_ports {led[5]}]
#set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33 } [get_ports {led[6]}]
#set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports {led[7]}]
#set_property -dict { PACKAGE_PIN V13   IOSTANDARD LVCMOS33 } [get_ports {led[8]}]
#set_property -dict { PACKAGE_PIN V3    IOSTANDARD LVCMOS33 } [get_ports {led[9]}]
#set_property -dict { PACKAGE_PIN W3    IOSTANDARD LVCMOS33 } [get_ports {led[10]}]
#set_property -dict { PACKAGE_PIN U3    IOSTANDARD LVCMOS33 } [get_ports {led[11]}]
#set_property -dict { PACKAGE_PIN P3    IOSTANDARD LVCMOS33 } [get_ports {led[12]}]
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#set_property -dict { PACKAGE_PIN N3    IOSTANDARD LVCMOS33 } [get_ports {led[13]}]
#set_property -dict { PACKAGE_PIN P1    IOSTANDARD LVCMOS33 } [get_ports {led[14]}]
#set_property -dict { PACKAGE_PIN L1    IOSTANDARD LVCMOS33 } [get_ports {led[15]}]


##7 Segment Display
#set_property -dict { PACKAGE_PIN W7   IOSTANDARD LVCMOS33 } [get_ports {seg[0]}]
#set_property -dict { PACKAGE_PIN W6   IOSTANDARD LVCMOS33 } [get_ports {seg[1]}]
#set_property -dict { PACKAGE_PIN U8   IOSTANDARD LVCMOS33 } [get_ports {seg[2]}]
#set_property -dict { PACKAGE_PIN V8   IOSTANDARD LVCMOS33 } [get_ports {seg[3]}]
#set_property -dict { PACKAGE_PIN U5   IOSTANDARD LVCMOS33 } [get_ports {seg[4]}]
#set_property -dict { PACKAGE_PIN V5   IOSTANDARD LVCMOS33 } [get_ports {seg[5]}]
#set_property -dict { PACKAGE_PIN U7   IOSTANDARD LVCMOS33 } [get_ports {seg[6]}]

#set_property -dict { PACKAGE_PIN V7   IOSTANDARD LVCMOS33 } [get_ports dp]

#set_property -dict { PACKAGE_PIN U2   IOSTANDARD LVCMOS33 } [get_ports {an[0]}]
#set_property -dict { PACKAGE_PIN U4   IOSTANDARD LVCMOS33 } [get_ports {an[1]}]
#set_property -dict { PACKAGE_PIN V4   IOSTANDARD LVCMOS33 } [get_ports {an[2]}]
#set_property -dict { PACKAGE_PIN W4   IOSTANDARD LVCMOS33 } [get_ports {an[3]}]


##Buttons
set_property -dict { PACKAGE_PIN U18   IOSTANDARD LVCMOS33 } [get_ports raw]
#set_property -dict { PACKAGE_PIN T18   IOSTANDARD LVCMOS33 } [get_ports btnU]
#set_property -dict { PACKAGE_PIN W19   IOSTANDARD LVCMOS33 } [get_ports btnL]
#set_property -dict { PACKAGE_PIN T17   IOSTANDARD LVCMOS33 } [get_ports btnR]
#set_property -dict { PACKAGE_PIN U17   IOSTANDARD LVCMOS33 } [get_ports btnD]


##Pmod Header JA
#set_property -dict { PACKAGE_PIN J1   IOSTANDARD LVCMOS33 } [get_ports {JA[0]}];#Sch
name = JA1
#set_property -dict { PACKAGE_PIN L2   IOSTANDARD LVCMOS33 } [get_ports {JA[1]}];#Sch
name = JA2
#set_property -dict { PACKAGE_PIN J2   IOSTANDARD LVCMOS33 } [get_ports {JA[2]}];#Sch
name = JA3
#set_property -dict { PACKAGE_PIN G2   IOSTANDARD LVCMOS33 } [get_ports {JA[3]}];#Sch
name = JA4
#set_property -dict { PACKAGE_PIN H1   IOSTANDARD LVCMOS33 } [get_ports {JA[4]}];#Sch
name = JA7
#set_property -dict { PACKAGE_PIN K2   IOSTANDARD LVCMOS33 } [get_ports {JA[5]}];#Sch
name = JA8
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#set_property -dict { PACKAGE_PIN H2   IOSTANDARD LVCMOS33 } [get_ports {JA[6]}];#Sch name = JA9
#set_property -dict { PACKAGE_PIN G3   IOSTANDARD LVCMOS33 } [get_ports {JA[7]}];#Sch name = JA10

##Pmod Header JB
#set_property -dict { PACKAGE_PIN A14   IOSTANDARD LVCMOS33 } [get_ports {JB[0]}];#Sch name = JB1
#set_property -dict { PACKAGE_PIN A16   IOSTANDARD LVCMOS33 } [get_ports {JB[1]}];#Sch name = JB2
#set_property -dict { PACKAGE_PIN B15   IOSTANDARD LVCMOS33 } [get_ports {JB[2]}];#Sch name = JB3
#set_property -dict { PACKAGE_PIN B16   IOSTANDARD LVCMOS33 } [get_ports {JB[3]}];#Sch name = JB4
#set_property -dict { PACKAGE_PIN A15   IOSTANDARD LVCMOS33 } [get_ports {JB[4]}];#Sch name = JB7
#set_property -dict { PACKAGE_PIN A17   IOSTANDARD LVCMOS33 } [get_ports {JB[5]}];#Sch name = JB8
#set_property -dict { PACKAGE_PIN C15   IOSTANDARD LVCMOS33 } [get_ports {JB[6]}];#Sch name = JB9
#set_property -dict { PACKAGE_PIN C16   IOSTANDARD LVCMOS33 } [get_ports {JB[7]}];#Sch name = JB10

##Pmod Header JC
#set_property -dict { PACKAGE_PIN K17   IOSTANDARD LVCMOS33 } [get_ports {JC[0]}];#Sch name = JC1
#set_property -dict { PACKAGE_PIN M18   IOSTANDARD LVCMOS33 } [get_ports {JC[1]}];#Sch name = JC2
#set_property -dict { PACKAGE_PIN N17   IOSTANDARD LVCMOS33 } [get_ports {JC[2]}];#Sch name = JC3
#set_property -dict { PACKAGE_PIN P18   IOSTANDARD LVCMOS33 } [get_ports {JC[3]}];#Sch name = JC4
#set_property -dict { PACKAGE_PIN L17   IOSTANDARD LVCMOS33 } [get_ports {JC[4]}];#Sch name = JC7
#set_property -dict { PACKAGE_PIN M19   IOSTANDARD LVCMOS33 } [get_ports {JC[5]}];#Sch name = JC8
#set_property -dict { PACKAGE_PIN P17   IOSTANDARD LVCMOS33 } [get_ports {JC[6]}];#Sch name = JC9
#set_property -dict { PACKAGE_PIN R18   IOSTANDARD LVCMOS33 } [get_ports {JC[7]}];#Sch name = JC10

##Pmod Header JXADC
#set_property -dict { PACKAGE_PIN J3   IOSTANDARD LVCMOS33 } [get_ports {JXADC[0]}];#Sch name = XA1_P
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#set_property -dict { PACKAGE_PIN L3   IOSTANDARD LVCMOS33 } [get_ports {JXADC[1]}];#Sch name = XA2_P
#set_property -dict { PACKAGE_PIN M2   IOSTANDARD LVCMOS33 } [get_ports {JXADC[2]}];#Sch name = XA3_P
#set_property -dict { PACKAGE_PIN N2   IOSTANDARD LVCMOS33 } [get_ports {JXADC[3]}];#Sch name = XA4_P
#set_property -dict { PACKAGE_PIN K3   IOSTANDARD LVCMOS33 } [get_ports {JXADC[4]}];#Sch name = XA1_N
#set_property -dict { PACKAGE_PIN M3   IOSTANDARD LVCMOS33 } [get_ports {JXADC[5]}];#Sch name = XA2_N
#set_property -dict { PACKAGE_PIN M1   IOSTANDARD LVCMOS33 } [get_ports {JXADC[6]}];#Sch name = XA3_N
#set_property -dict { PACKAGE_PIN N1   IOSTANDARD LVCMOS33 } [get_ports {JXADC[7]}];#Sch name = XA4_N


##VGA Connector
#set_property -dict { PACKAGE_PIN G19   IOSTANDARD LVCMOS33 } [get_ports {vgaRed[0]}]
#set_property -dict { PACKAGE_PIN H19   IOSTANDARD LVCMOS33 } [get_ports {vgaRed[1]}]
#set_property -dict { PACKAGE_PIN J19   IOSTANDARD LVCMOS33 } [get_ports {vgaRed[2]}]
#set_property -dict { PACKAGE_PIN N19   IOSTANDARD LVCMOS33 } [get_ports {vgaRed[3]}]
#set_property -dict { PACKAGE_PIN N18   IOSTANDARD LVCMOS33 } [get_ports {vgaBlue[0]}]
#set_property -dict { PACKAGE_PIN L18   IOSTANDARD LVCMOS33 } [get_ports {vgaBlue[1]}]
#set_property -dict { PACKAGE_PIN K18   IOSTANDARD LVCMOS33 } [get_ports {vgaBlue[2]}]
#set_property -dict { PACKAGE_PIN J18   IOSTANDARD LVCMOS33 } [get_ports {vgaBlue[3]}]
#set_property -dict { PACKAGE_PIN J17   IOSTANDARD LVCMOS33 } [get_ports {vgaGreen[0]}]
#set_property -dict { PACKAGE_PIN H17   IOSTANDARD LVCMOS33 } [get_ports {vgaGreen[1]}]
#set_property -dict { PACKAGE_PIN G17   IOSTANDARD LVCMOS33 } [get_ports {vgaGreen[2]}]
#set_property -dict { PACKAGE_PIN D17   IOSTANDARD LVCMOS33 } [get_ports {vgaGreen[3]}]
#set_property -dict { PACKAGE_PIN P19   IOSTANDARD LVCMOS33 } [get_ports Hsync]
#set_property -dict { PACKAGE_PIN R19   IOSTANDARD LVCMOS33 } [get_ports Vsync]


##USB-RS232 Interface
#set_property -dict { PACKAGE_PIN B18   IOSTANDARD LVCMOS33 } [get_ports RsRx]
#set_property -dict { PACKAGE_PIN A18   IOSTANDARD LVCMOS33 } [get_ports RsTx]


##USB HID (PS/2)
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#set_property -dict { PACKAGE_PIN C17   IOSTANDARD LVCMOS33   PULLUP true }
[get_ports PS2Clk]
#set_property -dict { PACKAGE_PIN B17   IOSTANDARD LVCMOS33   PULLUP true }
[get_ports PS2Data]


##Quad SPI Flash
##Note that CCLK_0 cannot be placed in 7 series devices. You can access it using the
##STARTUPE2 primitive.
#set_property -dict { PACKAGE_PIN D18   IOSTANDARD LVCMOS33 } [get_ports {QspiDB[0]}]
#set_property -dict { PACKAGE_PIN D19   IOSTANDARD LVCMOS33 } [get_ports {QspiDB[1]}]
#set_property -dict { PACKAGE_PIN G18   IOSTANDARD LVCMOS33 } [get_ports {QspiDB[2]}]
#set_property -dict { PACKAGE_PIN F18   IOSTANDARD LVCMOS33 } [get_ports {QspiDB[3]}]
#set_property -dict { PACKAGE_PIN K19   IOSTANDARD LVCMOS33 } [get_ports QspiCSn]


## Configuration options, can be used for all designs
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]

## SPI configuration mode options for QSPI boot, can be used for all designs
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]
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